Two-sample reading budget: same expert and context packing

CFPerm prototype

Not TMLE, not causal, not online learning. W is a batch tag; $\hat{e} \approx P(W{=}1 \mid X)$. Adaptation: freeze π from a domain pair, then feed a *single* reader the concatenated *raw* blocks $E = E(\pi)$. The reader is not updated from the query stream; π is not updated either. An opinion pool of $\hat{e}_m$ is a mixture of scores, not this map.

1 Raw block + one reader

Let $E = E(\pi)$. The observation is $\text{pack}(x) = \text{concat}_{m \in E}\text{serialize}(x_{B_m})$. Output is $f(\text{pack}(x))$ with one f. Dropped blocks are absent. This restricts the σ-algebra of a frozen reader. It is not Hedge, not a bandit over specialists, and not a per-query update of π.

2 Same expert versus packing

A *same expert* is $\hat{m} = \arg\max \pi$, then $f \in \mathcal{G}_{\hat{m}}$ (functions of $X_{B_{\hat{m}}}$ only) for every query. That template does not depend on x. An instance gate $\hat{m}(x) = \arg\max s_m(x)$ is a different object.

Context packing is one reader on concatenated raw blocks $E = E(\pi)$, not an opinion pool of $\hat{e}_m$. Disabled channels are omitted, not emitted as empty headers. Abstain means the gap is not localized: do not drop a block (pack all, or HITL). Forced top-1 on a diffuse shift discards signal.

3 Specialist $\hat{e}_m$ (one block, all rows)

4 Packed reader (one RF on concat E)

Inject valence with a wide text block: VIMP hires the wide specialist; π hires valence and the packed reader matches the GT block. Diffuse equal shift: pack-all beats one specialist. Shuffle valence: hit collapses.

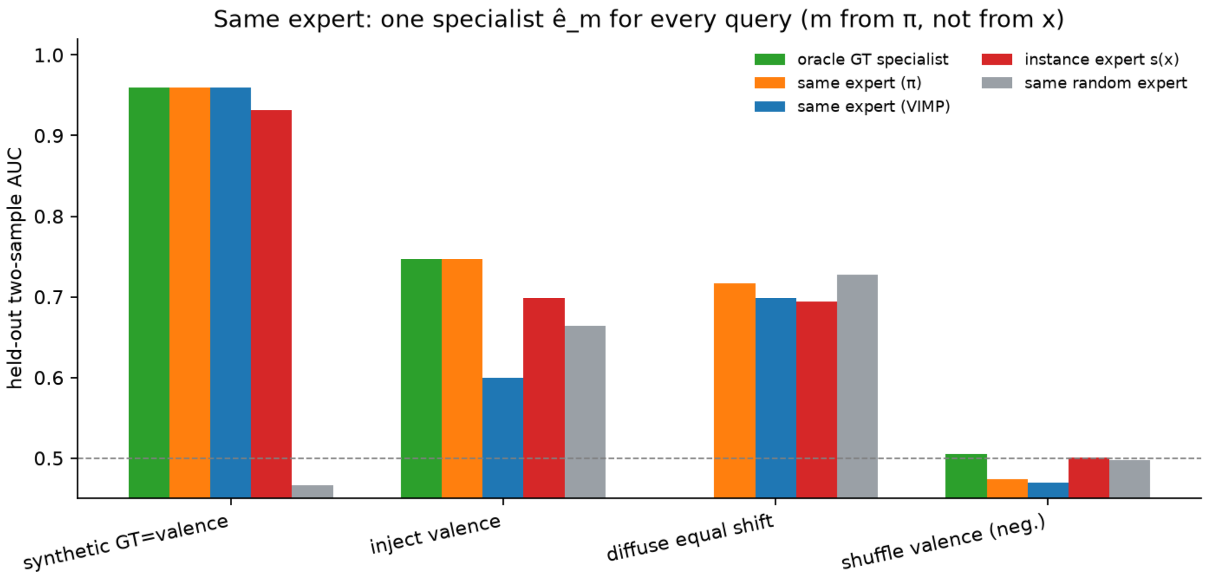

Figure 1: Frozen specialist AUC. π picks one m for the whole test set.

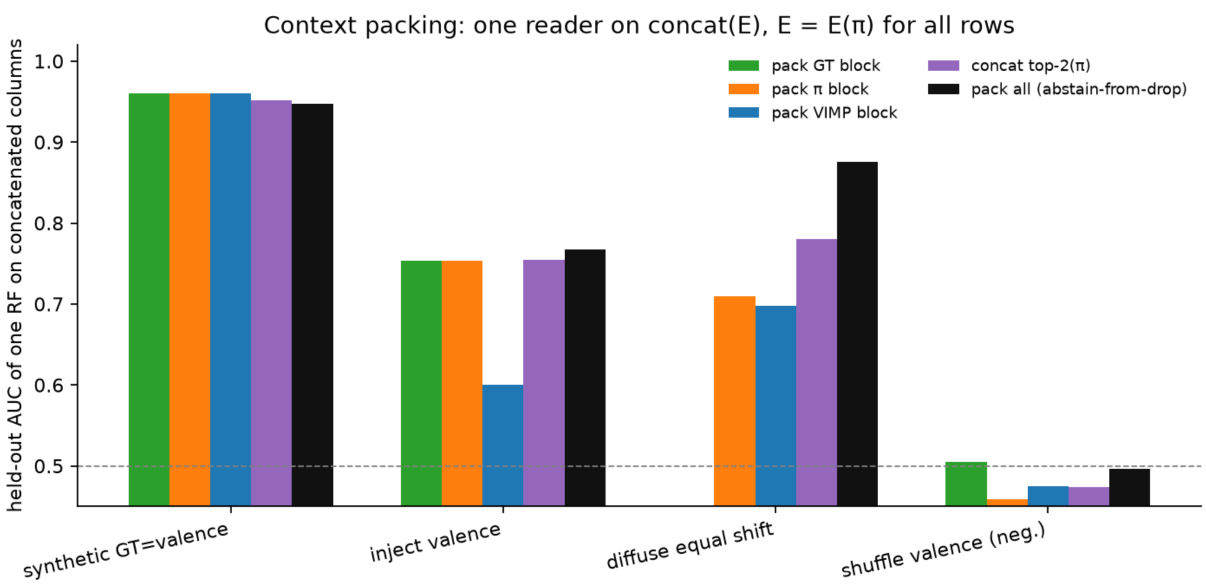

Figure 2: Column-concat of $E(\pi)$. Pack-all is the abstain-from-dropping action.

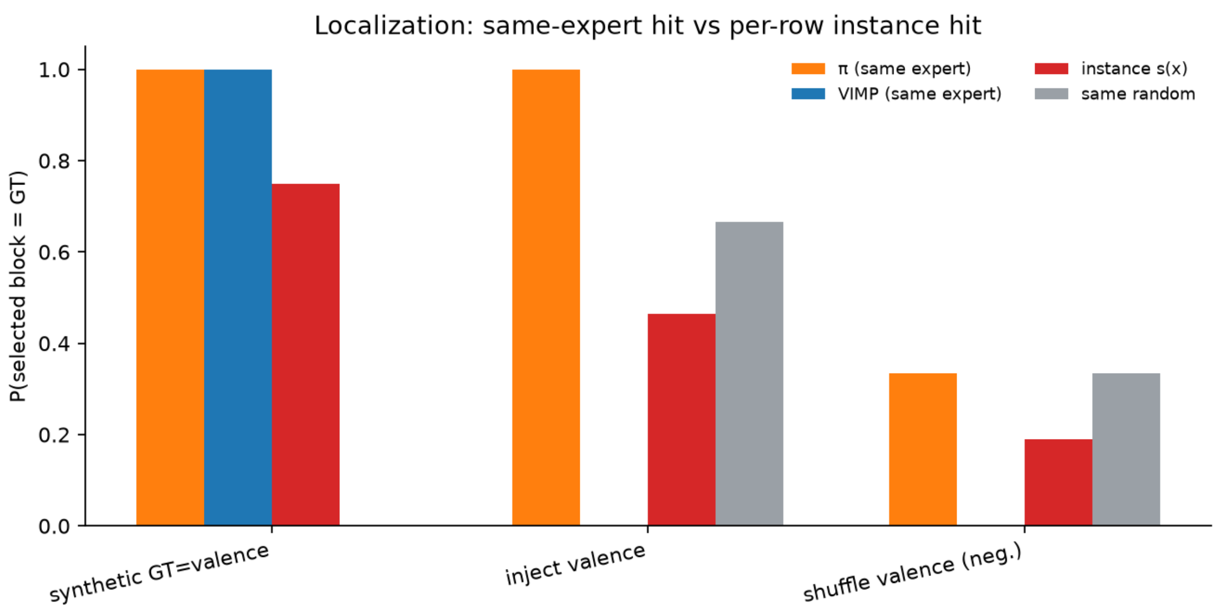

Figure 3: Localization hit. Same-expert hit is $1\{\hat{m} = m^\star\}$, not a per-row mix.